Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	29
interior lacuna count	25
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 25)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2817.72	1943	2635.45	px^2
major_axis_length_px	lacuna shape	83.2391	74.781	32.4447	px
minor_axis_length_px	lacuna shape	44.1848	37.2098	22.4578	px
aspect_ratio	lacuna shape	2.0619	2.0239	0.6289	unitless
eccentricity	lacuna shape	0.8262	0.8694	0.1152	unitless
solidity	lacuna shape	0.7461	0.7383	0.0965	unitless
roots_count	headline	10.96	10	5.7986	count
ring_length_r30_px	headline	291.12	277	139.638	px
ring_length_r60_px	reported	568.52	585	287.782	px
owned_length_px	ownership-dependent	663.818	437.676	608.023	px
edge_count	ownership-dependent	33.2	23	29.1305	count
mean_edge_length_px	ownership-dependent	20.0158	19.6399	6.2126	px
perimeter_px	normalised	273.398	253.279	124.13	px
ring_area_r30_px2	normalised	9128.76	8169	3446.34	px^2
ring_area_r60_px2	normalised	18617.1	17028	6982.69	px^2
in_frame_fraction_r30	normalised	0.987	1	0.0329	unitless
in_frame_fraction_r60	normalised	0.9657	1	0.0725	unitless
ring_density_r30	normalised	0.0318174	0.0315741	0.0101964	px^-1
ring_density_r60	normalised	0.0302591	0.0316249	0.0101404	px^-1
roots_per_100px_perimeter	normalised	4.0036	4.137	1.0972	per 100 px
ring_attached_length_r30_px	network v2	263.36	232	142.18	px
ring_attached_length_r60_px	network v2	440.2	357	294.358	px
ring_length_w_r30_px	network v2	330.216	314.907	158.203	px
ring_length_w_r60_px	network v2	644.577	660.688	327.258	px
sholl_crossings_r10	network v2	6.6	6	3.4278	count
sholl_crossings_r20	network v2	7.32	7	4.0489	count
sholl_crossings_r30	network v2	8.16	7	4.4317	count
loop_count	loops, report only	1.68	0	2.7343	count
loop_edge_share	loops, report only	0.1088	0	0.1445	share
ring_width_mean_r30_px	width	7.8121	7.6247	1.2378	px

Per field
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measure	value	units
analysed_area_px2	771553	px^2
total_skeleton_length_px	22883	px
canalicular_length_density_per_px	0.0296584	px^-1
junction_count	507	count
junction_density_per_px2	0.00065712	px^-2
lacuna_count	29	count

lacunae_per_px2	3.759e-05	px^-2
skeleton_component_count	270	count
mean_component_length_px	84.7519	px
median_component_length_px	22.5	px
field_length_density_w_per_px	0.0336573	px^-1
field_density_without_flagged_per_px	0.0237471	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	158	px
unattached_component_count	11	count
canalicular_length_density_attached_per_px	0.0296584	px^-1
field_length_density_with_unattached_per_px	0.0298632	px^-1
excluded_area_px2	193353	px^2
analysed_field_area_px2	855223	px^2
loop_count	55	count
loop_edge_share	0.1424	share
edge_count	1390	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2098	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23450	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	1943	78.2261	36.7629	2.1279	0.8827	0.7692	-1.4856	56.1004	636.36	False
4	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
5	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
6	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
7	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
8	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
9	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
10	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
11	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
12	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
13	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
14	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
15	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
16	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
17	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
18	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
19	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
20	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
21	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
22	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False

23	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False
24	549	48.7614	15.8873	3.0692	0.9454	0.8356	-0.9101	738.328	786.444	False
25	640	57.661	17.8443	3.2313	0.9509	0.7018	-0.3969	793.93	358.258	False
26	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
27	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False
28	6240	115.676	84.3355	1.3716	0.6844	0.7226	-1.0938	881.946	274.251	False
29	1277	51.634	38.2716	1.3491	0.6713	0.7854	1.2894	872.15	1003.8	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	337	443	568.174	29	19.5922	6.7223
2	11	264	350	288.108	21	13.7194	6.4414
3	13	309	613	437.676	31	14.1186	7.2615
4	9	352	597	848.37	50	16.9674	7.3539
5	10	277	601	177.853	13	13.681	6.6277
6	15	438	695	792.855	39	20.3296	7.5509
7	8	322	799	783.512	40	19.5878	7.7452
8	6	155	358	401.421	19	21.1274	8.7355
9	26	659	1235	2350.11	115	20.4357	8.1509
10	7	199	419	426.262	23	18.5331	8.5102
11	16	500	1055	2184.78	102	21.4194	8.5149
12	11	341	658	1054.25	55	19.1681	8.3777
13	8	192	330	232.836	10	23.2836	6.0936
14	12	360	668	1001.49	35	28.614	7.7108
15	5	155	258	175.125	10	17.5125	8.2917
16	19	351	508	723.027	34	21.2655	5.9354
17	19	385	862	1396.18	70	19.9455	7.8658
18	14	408	1030	1085.1	51	21.2766	7.1052
19	15	277	585	667.245	36	18.5346	7.6247
20	9	167	339	207.48	7	29.64	7.1073
21	23	499	869	903.436	46	19.6399	7.5211
22	9	207	349	267.61	17	15.7418	7.6147
23	5	167	317	118.498	6	19.7496	7.8453
24	4	187	407	201.723	5	40.3446	7.5441
25	1	35	62	7.8284	1	7.8284	12.9591
26	8	232	483	506.031	22	23.0014	8.1991
27	6	119	213	157.468	10	15.7468	7.2164
28	10	227	419	323.622	15	21.5748	8.0692
29	5	149	482	665.76	38	17.52	10.4527

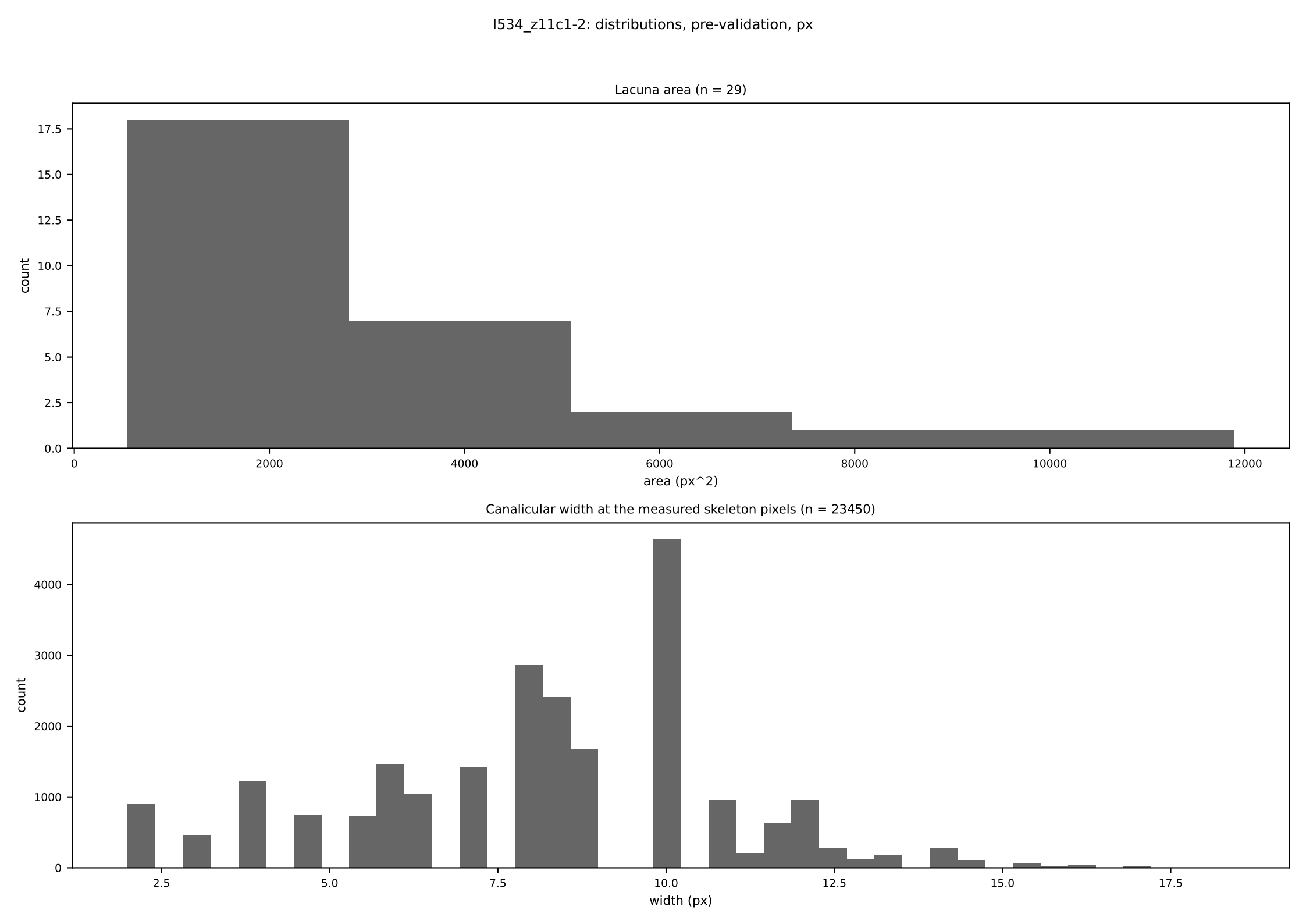
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	12179	0.7421	0.6708	0.0409428	0.0363741	3.0343
2	282.291	4862	7142	0.9455	0.8134	0.0542986	0.0490059	3.8967
3	253.279	9149	18014	1	0.9073	0.0337742	0.0340291	5.1327
4	270.635	8169	16049	0.8695	0.7737	0.0430897	0.0371986	3.3255
5	236.936	6970	15047	1	1	0.0397417	0.0399415	4.2205
6	341.505	11247	22740	1	1	0.0389437	0.0305629	4.3923
7	176.167	7690	19162	1	1	0.0418726	0.0416971	4.5412
8	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
9	548.257	17191	33534	1	1	0.038334	0.0368283	4.7423
10	136.125	6663	16736	1	1	0.0298664	0.0250358	5.1423
11	297.706	11030	23297	1	1	0.0453309	0.0452848	5.3744
12	305.605	10599	17028	1	1	0.0321729	0.0386422	3.5994
13	191.167	7524	16666	0.9189	0.7962	0.0255183	0.0198008	4.1848

14	238.451	9266	19320	1	1	0.0388517	0.0345756	5.0325
15	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
16	490.399	11727	20563	0.8061	0.7376	0.0299309	0.0247046	3.8744
17	378.475	12733	27257	1	1	0.0302364	0.0316249	5.0201
18	373.546	12922	30033	1	1	0.0315741	0.0342956	3.7479
19	290.877	7526	16187	1	1	0.0368057	0.0361401	5.1568
20	233.38	6410	12080	1	1	0.026053	0.0280629	3.8564
21	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
22	383.889	12892	25468	1	1	0.0160565	0.0137035	2.3444
23	97.9828	5438	10968	1	1	0.0307098	0.0289023	5.1029
24	115.539	5958	13933	1	1	0.0313864	0.0292112	3.462
25	139.054	4710	7157	1	1	0.007431	0.00866285	0.7191
26	222.693	8786	20073	1	0.9917	0.0264056	0.0240622	3.5924
27	196.066	6012	12715	1	1	0.0197937	0.0167519	3.0602
28	446.416	13647	28597	1	1	0.0166337	0.0146519	2.2401
29	206.752	5125	12818	0.6755	0.6216	0.0290732	0.0376034	2.4184

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	334	421	381.806	497.647	9	8	4
2	257	292	296.279	394.706	9	5	3
3	210	357	347.735	683.63	5	7	11
4	348	545	392.836	660.688	6	9	6
5	179	179	314.907	678.257	6	4	8
6	418	618	503.274	794.796	11	11	13
7	284	620	367.563	917.051	6	11	10
8	140	252	165.669	393.836	3	4	4
9	639	1187	726.487	1384.67	15	15	17
10	192	283	230.137	479.375	5	6	3
11	486	1025	568.688	1201.18	10	16	18
12	327	612	395.505	761.553	8	9	12
13	148	172	222.936	387.103	5	2	5
14	340	593	412.605	776.039	6	7	9
15	145	178	175.297	287.551	4	4	4
16	327	352	403.576	592.541	7	3	5
17	376	722	438.233	975.264	11	10	11
18	395	767	467.961	1158.28	8	11	12
19	232	384	316.179	665.186	3	6	7
20	153	183	181.853	382.019	6	3	3
21	451	748	578.399	1005.9	13	14	13
22	199	281	226.853	381.007	5	4	7
23	148	244	184.468	356.735	4	6	4
24	115	165	215.066	467.031	4	7	7
25	10	10	37.799	67.6985	0	1	2
26	232	429	267.794	549.902	6	6	7
27	119	159	135.397	236.238	2	2	3
28	181	252	251.137	462.576	7	7	9
29	143	409	162.468	544.517	5	5	4


I534_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 29)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 23450)
count
width (px)